Lecture 2: PageRank and Random Walks

1 Ranking Webpages

Suppose a search engine finds one million pages containing the word `university`. Which pages should it display first?

Text matching tells us whether a page mentions a topic. Links provide another kind of information: which pages other pages refer to.

A first idea: count incoming links Why might a page with many incoming links be useful? Can you think of a way to make this rule give a misleading ranking?

Not all links should count equally PageRank uses two ideas:

- A link from an important page should contribute more.
- More outgoing links divide a page's contribution among more destinations.

The difficulty is that we seem to need to know which pages are important before we can determine which pages are important.

Question 1. How could we get started without knowing the final scores?

(a) Give every page an initial score and repeatedly update the scores

(b) Choose the most important page in advance

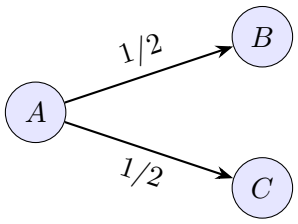
(c) Ignore all outgoing links

We will represent the web as a directed graph, define an update rule, and analyze an algorithm that applies that rule efficiently.

2 A Random Surfer

Imagine a person browsing the web. At each page, the person chooses one outgoing link uniformly at random and follows it.

A local rule If a page has four outgoing links, the probability of following any particular one is 1/4. To choose the next page, the surfer only needs to know the current page and its outgoing links.



The Markov property A **Markov chain** is a random process in which the next-state probabilities depend only on the current state. They do not require the entire history of how we arrived there.

For our surfer, the states are webpages. We assume that the graph and the transition rule stay fixed while we analyze the walk.

Question 2. Two surfers reach the same page by different routes. Under our rule, do they have the same probabilities for their next page?

(a) Yes

(b) No, because their browsing histories differ

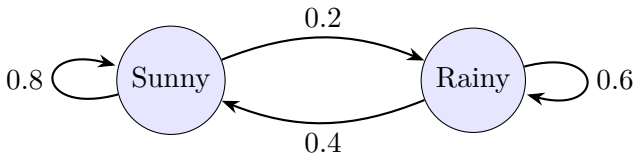
The ranking idea is to give a high score to a page that the surfer visits often. First, we need a way to keep track of the surfer's probabilities.

A small Markov-chain example

Suppose we use the following simplified model of weather:

- After a sunny day, tomorrow is sunny with probability 0.8.
- After a rainy day, tomorrow is sunny with probability 0.4.

This is a model for the example, not a claim that real weather depends only on whether today is sunny or rainy.



Predict one day ahead If today is sunny, tomorrow's probabilities are

$$\Pr[\text{sunny tomorrow}] = 0.8, \qquad \Pr[\text{rainy tomorrow}] = 0.2.$$

Predict two days ahead There are two ways to be sunny two days from now:

- Sunny tomorrow, then sunny again: probability $0.8 \cdot 0.8$.
- Rainy tomorrow, then sunny: probability $0.2 \cdot 0.4$.

Adding gives

$$\Pr[\text{sunny in two days}] = 0.72.$$

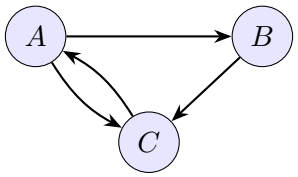
What is the probability of rain in two days?

Why is this useful? A simple rule for one step lets us predict later behavior. Similar models describe movement through a network, customer queues, and machine operating states, when the chosen state includes the information needed for the next step.

For PageRank, we will perform the same kind of calculation: multiply along each possible transition, then add the contributions arriving at a page.

3 Distributing Scores Along Links

We will use the following graph throughout the lecture:



Let $p(u)$ be the probability that the surfer is currently at page u. The probabilities sum to 1. We also call these values the current scores.

First, decide what each page sends Write $\deg^+(u)$ for the number of outgoing links from u. Each outgoing link receives a contribution of

$$\frac{p(u)}{\deg^+(u)}.$$

For now, every page has at least one outgoing link.

Next, add what each page receives Start with $p(A) = p(B) = p(C) = 1/3$. Fill in the new scores:

$$p_{\text{new}}(A) = 1/3, \qquad p_{\text{new}}(B) = 1/6, \qquad p_{\text{new}}(C) = 1/2.$$

Check that the new scores still sum to 1. Then perform one more round.

All contributions in a round must use the old scores. We should not use a newly computed score until the next round.

When have the scores settled down?

The link-following update is

$$p_{\text{new}}(v) = \sum_{u:u\to v} \frac{p(u)}{\deg^+(u)}.$$

A set of probabilities that is unchanged by this update is called a **stationary distribution**.

Check a proposed answer For our three-page graph, suppose

$$p(A) = \frac{2}{5}, \qquad p(B) = \frac{1}{5}, \qquad p(C) = \frac{2}{5}.$$

Without solving any equations, check what happens after another round:

- A receives $2/5$ from C.
- B receives $1/5$ from A.
- C receives $1/5 + 1/5 = 2/5$ from A and B.

The probabilities are unchanged, even though the surfer continues moving.

Question 3. Pages A and B each have one incoming link. Must they have the same score?

(a) Yes, the number of incoming links determines the score

(b) No, the contributions depend on which pages link to them

What does stationary mean? Does it mean that the surfer eventually stops at one page?

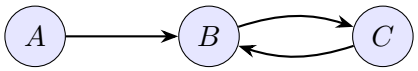
This example has sensible stationary scores. Unfortunately, following links alone does not always give a useful ranking or an iteration that settles down.

4 Why Following Links Is Not Enough

Dead ends Suppose the only link is $A \to B$, and B has no outgoing links. What should the surfer do after reaching B?

In score updates, simply discarding the score at B would make the total score smaller than 1.

Spider traps Now consider this graph:



Once the surfer enters $\{B, C\}$, it cannot leave. A closed group of pages like this is called a **spider trap**.

Why might a group receive too much importance just because it links only to itself?

Scores can also oscillate With just two links $A \to B$ and $B \to A$, start with scores $(1, 0)$. What are the next three pairs of scores?

We need a rule that lets the surfer leave a page or a group of pages without depending entirely on the available links.

5 PageRank: Add Random Jumps

Choose a number d with $0 < d < 1$. At a page that has outgoing links:

- With probability d, follow a uniformly chosen outgoing link.
- With probability $1 - d$, jump to a uniformly chosen page.

At a page with no outgoing links, always jump to a uniformly chosen page. A jump may choose the current page again.

The random-surfer interpretation The jumps are called **teleportation**. For example, if $d = 0.85$, the surfer follows a link with probability 0.85 and chooses a random page with probability 0.15, when outgoing links exist.

Question 4. Why can a surfer now escape a spider trap?

(a) We delete the links inside the trap

(b) A random jump can move to a page outside the trap

(c) Every trapped page must gain a new ordinary link

The update when there are no dead ends If there are n pages, then

$$p_{\text{new}}(v) = \frac{1-d}{n} + d \sum_{u:u\to v} \frac{p(u)}{\deg^+(u)}.$$

The first term is the contribution from random jumps. The second is the contribution from following links.

What we can guarantee With these jumps and the dead-end rule, there is a unique stationary set of scores. Repeated updates approach those scores from any starting probability distribution. We will use this fact without proving it.

These stationary scores are the **PageRank scores**. They describe how often the modified random surfer visits each page in the long run.

Compute two PageRank rounds

Return to the links $A \to B$, $A \to C$, $B \to C$, and $C \to A$. Use $d = 1/2$ to keep the arithmetic simple, and start with score $1/3$ at each page.

First, add the jump contribution There are three pages, so each receives

$$\frac{1-d}{3} = \frac{1}{6}.$$

Next, add the link contributions For this graph, one round is

$$\begin{aligned} p_{\text{new}}(A) &= \frac{1}{6} + \frac{1}{2}p(C), \\ p_{\text{new}}(B) &= \frac{1}{6} + \frac{1}{4}p(A), \\ p_{\text{new}}(C) &= \frac{1}{6} + \frac{1}{4}p(A) + \frac{1}{2}p(B). \end{aligned}$$

Why is the coefficient of $p(A)$ equal to $1/4$ rather than $1/2$?

Fill in the table

Round	A	B	C
0	1/3	1/3	1/3
1	1/3	1/4	5/12
2	3/8	1/4	3/8

Check that each row sums to 1. Which page is ranked first after each round?

After many rounds, the scores approach approximately $(0.359, 0.256, 0.385)$, so the final ordering is $C > A > B$. A small number of rounds need not give the final ordering, especially when scores are close.

6 Implementing the Algorithm

We can compute all the new scores by scanning the outgoing links of each page. Store the graph using adjacency lists.

Remember the pages with no outgoing links Let z be the sum of their current scores. In addition to the ordinary jump contribution, distribute their remaining score dz uniformly. Thus each page starts the round with $((1-d)+dz)/n$.

PageRank(G, d, K), where $0 < d < 1$ and $K \geq 1$

Let $n = |V|$ and compute the out-degree $\deg^+(u)$ of every page
Initialize $p(u) \leftarrow 1/n$ for every page u
for $t = 1, \ldots, K$ **do**
  $z \leftarrow \sum_{u:\deg^+(u)=0} p(u)$
  $p_{\text{new}}(v) \leftarrow ((1-d)+dz)/n$ for every page v
  for each edge $(u, v) \in E$ **do**
    $p_{\text{new}}(v) \leftarrow p_{\text{new}}(v) + d\,p(u)/\deg^+(u)$
  end for
  $p \leftarrow p_{\text{new}}$
end for
return p

Why use two arrays? Why should we not overwrite $p(v)$ immediately as we scan the edges?

Question 5. What is the running time of one round?

(a) $O(n + |E|)$

(b) $O(n^2)$, because a jump can go to any page

(c) $O(|E|^2)$

With K rounds, the total running time is $O(K(n+|E|))$. The two score arrays use $O(n)$ space in addition to the graph. The returned scores after K rounds are approximations.

Check the dead-end rule

Suppose there are only two pages, with the single link $A \to B$. Page B has no outgoing links. Start with $p(A) = p(B) = 1/2$, and use $d = 1/2$.

Follow what happens to the score at A Half of A's score follows its link to B. The other half is split equally between the two pages by random jumps.

$$\text{From } A \text{ to } A\text{: } \frac{1}{8}, \qquad \text{from } A \text{ to } B\text{: } \frac{3}{8}.$$

Follow what happens to the score at B There is no outgoing link, so all of B's score is distributed uniformly:

$$\text{From } B \text{ to } A\text{: } \frac{1}{4}, \qquad \text{from } B \text{ to } B\text{: } \frac{1}{4}.$$

Add the contributions The new scores are

$$p_{\text{new}}(A) = \frac{3}{8}, \qquad p_{\text{new}}(B) = \frac{5}{8}.$$

Check that their sum is 1.

Now compare with the algorithm Here $z = 1/2$, so the initialization gives every page

$$\frac{(1-d) + dz}{2} = \frac{3}{8}.$$

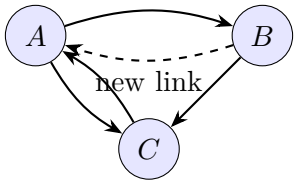
The edge $A \to B$ then adds $1/4$ to B.

Question 6. What would go wrong if we initialized each score to $(1-d)/2$ and then scanned the edges, without accounting for z?

Every page must distribute all of its score. Having no outgoing links is a case the implementation must handle explicitly.

7 What Happens If We Add a Link?

Work with a partner. Start with our original three-page graph, then add the link $B \to A$. The new graph is shown below.



Use $d = 1/2$ and restart the calculation with score $1/3$ at each page.

Predict before calculating Which page receives a new contribution? Which old contribution becomes smaller?

Compute one round

$$p_{\text{new}}(A) = \frac{5}{12}, \qquad p_{\text{new}}(B) = \frac{1}{4}, \qquad p_{\text{new}}(C) = \frac{1}{3}.$$

Compare this with the old first-round scores $(1/3, 1/4, 5/12)$.

Explain the change Page C still has the same two incoming links. Why did its score decrease in this round?

This compares one round from the same starting scores. To find the new stationary ranking, we would continue applying updates on the new graph.

8 Using and Interpreting PageRank

A score for the graph Ordinary PageRank depends on the link graph, not on a particular search query. We can compute it in advance and combine it with information about how well each page matches a query.

Would a page with a high PageRank necessarily be a good answer to a search for a particular restaurant's opening hours?

Changing the jump rule For **personalized PageRank**, random jumps favor selected pages, such as pages about a topic or items a user likes. Following links then spreads that emphasis through the graph.

What might this be useful for besides ranking webpages?

Changing d More frequent jumps make the starting point matter less and put more emphasis on the jump rule. More link-following makes the graph structure more influential.

If $d = 0$ and jumps are uniform, what scores do we obtain?

Question 7. Which statement best describes PageRank?

(a) It counts incoming links

(b) It computes scores from a random-surfer model by repeated updates

(c) It guarantees that highly ranked pages are true and useful

The main algorithmic ideas are to model a problem using a graph, define a local update, and compute repeated updates by scanning vertices and edges.